

Safety Belt System



Principles of Operation

WARNING: All seatbelt components must be inspected and corrected as part of any collision repair. Inspect all seatbelt components as prescribed by Seatbelt Procedure After a Collision found in Section 501-20A General Procedures of this manual. Failure to follow this instruction may result in incorrect operation of the seatbelt system and increase the risk of serious personal injury or death in a crash.

Safety Belt Warning System

- The conditions of operation for the driver and/or passenger safety belt warning indicator and chime operate as follows:
- If the driver and/or passenger safety belt is not buckled with the ignition ON, then the safety belt warning indicator illuminates for 1 to 2 minutes and the warning chime sounds for 4 to 8 seconds
 - If the driver and/or passenger safety belt is buckled while the warning indicator is illuminated and the warning chime is sounding, then the indicator and chime will turn off
 - If the driver and/or passenger safety belt is buckled before the ignition is in the ON position, the safety belt warning indicator and chime will remain off

Belt-Minder®

The Belt-Minder® feature is a supplemental warning to the safety belt warning function. This feature provides an additional reminder that the driver and/or front passenger safety belt is unbuckled by intermittently sounding a chime and illuminating the safety belt warning indicator in the Instrument Panel Cluster (IPC) .

Either the driver or front passenger safety belt systems may activate the Belt-Minder®. If the Belt-Minder® warnings have expired (warnings for approximately 5 minutes) for one occupant (driver or front passenger), the other occupant can still activate the Belt-Minder® feature.

When a MyKey® is in the ignition, deactivation of the Belt-Minder® is ignored. The Belt-Minder® feature can be activated or deactivated by the administrator key. Using the non-administrator key overrides the deactivation of the Belt-Minder® feature. Refer to Section 419-01B or the Owner's Literature for further information on MyKey® functionality.

The passenger Belt-Minder® feature is activated only when the Occupant Classification System (OCS) system detects an occupant in the right front seat and the occupants weight exceeds a programmed limit. For information on the OCS system, refer to Section 501-20B.

To activate or deactivate the Belt-Minder® feature, refer to Section 413-01.

If...	Then...
the driver and/or passenger safety belt is buckled before the ignition is in the ON position or less than 1-2 minutes have elapsed since the ignition has been turned ON...	the Belt-Minder® feature does not activate.
the driver and/or front passenger safety belt is not buckled before the vehicle has reached at least 5 km/h (3 mph) and	the Belt-Minder® feature is activated — the safety belt warning indicator illuminates and the warning chime sounds

If...	Then...
1-2 minutes have elapsed since the ignition has been turned ON...	for 6 seconds every 30 seconds, repeating for approximately 5 minutes or until the safety belt(s) is buckled.
the driver or front passenger safety belt becomes unbuckled for approximately one minute while the vehicle is traveling at least 5 km/h (3 mph) and more than 1-2 minutes have elapsed since the ignition has been ON...	the Belt-Minder® feature is activated — the safety belt warning indicator illuminates and the warning chime sounds for 6 seconds every 30 seconds, repeating for approximately 5 minutes or until the safety belt(s) is buckled.

Inspection and Verification

1. Verify the customer concern.
2. Inspect to determine if any of the following mechanical or electrical concerns apply:

Visual Inspection Chart

Mechanical	Electrical
• Safety belt webbing integrity (torn, frayed, cut or stretched) • Safety belt buckle and tongue assembly • Safety belt retractor	• Bare, broken or disconnected wire • Connector not tightly engaged • Safety belt warning indicator burned out or broken

3. If the inspection reveals an obvious concern(s) that can be readily identified, service as required. With the exception of removing a twist from the safety belt webbing, do not attempt to repair a component of the safety belt system; new components must be installed.
4. If the concern remains after the inspection, determine the symptom. GO to [Symptom Chart](#).
5. To check the active restraint system for correct operation, carry out the appropriate Functional Test procedure(s). Refer to Component Test in this section.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
• The safety belt warning indicator does not operate	• Burned-out indicator/LED • Wiring, terminals or connectors • Instrument Panel Cluster (IPC) • Restraints Control Module (RCM)	• If the air bag warning indicator is illuminated, REFER to Section 501-20B. If the air bag warning indicator is not illuminated, REFER to Section 413-01.

• The safety belt warning chime does not operate	• Wiring, terminals or connectors • Safety belt switch • IPC • RCM	• If the air bag warning indicator is illuminated, REFER to Section 501-20B. If the air bag warning indicator is not illuminated, REFER to Section 413-01.
• The safety belt retractor webbing does not extract or exhibits excessive pressure on the occupant during use	• Safety belt retractor is in web sensing mode • Interior trim is not installed correctly • Safety belt retractor is in Automatic Locking Retractor (ALR) mode • Safety belt retractor is binding internally	• UNLOCK the locked safety belt retractor. REFER to Locked Safety Belt Retractor Releasing in this section. • CORRECT the interior trim installation. REFER to Section 501-05. • CARRY OUT the Functional Test — Safety Belt Retractor, Functional Test — Safety Belt Retractor, Automatic Locking Retractor (ALR) Mode or Functional Test - Safety Belt Retractor, Road Test Inspection, as appropriate. REFER to Component Test in this section. INSTALL new components as directed.
• The safety belt retractor webbing does not retract or has excessive slack during use	• Twisted or incorrectly routed safety belt webbing • Interior trim is not installed correctly • Safety belt retractor is binding internally	• CORRECT the twisted safety belt webbing. REFER to Safety Belt Twisted at the Safety Belt Guide or Safety Belt Tongue Rotated on Belt in this section. CARRY OUT the appropriate Functional Test. REFER to Component Test in this section. • CORRECT the interior trim installation. REFER to Section 501-05. • CARRY OUT the Functional Test — Safety Belt Retractor, Functional Test — Safety Belt Retractor, Automatic Locking Retractor (ALR) Mode or Functional Test - Safety Belt Retractor, Road Test Inspection, as appropriate. REFER to Component Test in this section. INSTALL new components as directed.
• The Belt-Minder® feature is inoperative	• Wiring, terminals or connectors • Belt-Minder® feature is deactivated	• REFER to Section 413-01 or the Seating and Safety Restraints section of the Owner's Literature to activate the Belt-Minder® feature. If the Belt-Minder® is activated, REFER to Section 413-01 to diagnose.

• The safety belt tongue is rotated on the safety belt webbing	• Twisted or incorrectly routed safety belt webbing	• REFER to Safety Belt Tongue Rotated on Belt in this section.
• The safety belt webbing is twisted at the safety belt guide	• Twisted or incorrectly routed safety belt webbing	• REFER to Safety Belt Twisted at the Safety Belt Guide in this section.
• The safety belt webbing is dirty or soiled	• Dirty or soiled safety belt webbing	• REFER to Safety Belt Cleaning in this section.

Component Test

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Carry out the appropriate Functional Test(s) as determined in Inspection and Verification.

Functional Test — Safety Belt Buckle and Tongue

The safety belt buckle and tongue assembly must operate freely during the latching and unlatching function. Fasten the safety belt by inserting the tongue (male portion) into the buckle (female portion).

1. Verify the following during the latching sequence:
 - Tongue insertion is not hindered by excessive effort.
 - A click is heard when the safety belt buckle latches the tongue.
2. Verify the system integrity by forcefully pulling on the safety belt webbing.
3. Unlatch the safety belt by fully depressing the safety belt buckle release button and allowing the safety belt to release and retract.
4. Verify the following during the unlatching process:
 - Push-button depression does not require excessive effort.
 - The tongue can be removed easily from the buckle.
5. Repeat the above steps 3 times.
6. If the functional test reveals a concern(s), install a new safety belt buckle or safety belt retractor as required.

Functional Test — Safety Belt Retractor

The safety belt retractor must be freely operational for extraction and retraction of the safety belt webbing between full extension and in-vehicle stowed positions.

1. Extract and retract the safety belt between the full extension and stowed positions.
2. Verify the safety belt retractor operates without excessive effort or binding.
3. Install a new safety belt retractor if any concern is found or the complaint has been verified.

Functional Test — Safety Belt Retractor, Automatic Locking Retractor (ALR) Mode

1. Position the seat backrest fully upright (if adjustable).

2. Position the safety belt shoulder height adjuster (if equipped) in the full down or up position.
3. Fasten the safety belt.
4. Pull out the safety belt webbing until the Automatic Locking Retractor (ALR) feature is activated.
5. Allow the safety belt webbing to retract until it stops.
6. Pull on the safety belt webbing to check that the safety belt retractor has remained in the ALR mode. If the safety belt retractor is not locked, install a new safety belt retractor.
7. Unfasten the safety belt and allow the safety belt webbing to retract to its stowed position.
8. Pull out and retract the safety belt webbing to verify the safety belt retractor has converted automatically out of ALR mode. If the safety belt retractor remains locked in the ALR mode, install a new safety belt retractor.

Functional Test — Safety Belt Retractor, Road Test Inspection

WARNING: The driver and passenger must be prepared to brace themselves in the event the seatbelt retractor does not lock. Failure to follow this instruction may result in serious personal injury.

NOTE: *Make sure there is no excessive slack in the safety belt webbing across the torso during testing.*

NOTE: *Do not jerk on the safety belt webbing when carrying out this test.*

1. Test the safety belts in the following sequence:
 1. Fasten the safety belts and proceed to a safe area.
 2. Attain a speed of 24 km/h (15 mph).
3. **WARNING: Apply maximum brake force only on dry concrete or equivalent hard surface, never on wet pavement or gravel. Failure to follow this instruction may result in serious personal injury.**

Grasp the shoulder harness, lean forward and apply the brakes, making a maximum braking application without a skid.
4. The safety belts should lock up with minimum webbing extension.
5. If there is a lockup of all safety belt retractors being tested, the safety belt retractors are functioning correctly. If any safety belt retractor fails to lock up, install a new safety belt retractor(s).